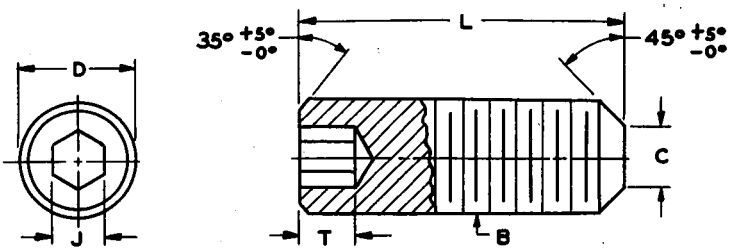
FED. SUP. CLASS.
5305



D	NOMINAL SIZE		#0 (.060)		#2 (.086)		#4 (.112)		#6 (.138)		#8 (.164)	
B	THREADS PER INCH		80NF-3A		64NF-3A		48NF-3A		40NF-3A		36NF-3A	
J	WIDTH OF SOCKET ACROSS FLAT	MAX.	.0285		.0355		.051		.0635		.0791	
		MIN.	.028		.035		.050		.0625		.0781	
C	DIAMETER OF POINT	MAX.	.033		.047		.061		.074		.087	
		MIN.	.027		.039		.051		.064		.076	
T	KEY ENGAGEMENT	MIN.	.022		.028		.040		.050		.062	
L	LENGTH	TOLERANCE	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN
1/16		± 1-1/2% OF L WITH A MINIMUM OF ± .010 IN.	1*									
1/8					7*		16*		25*		35*	
3/16					8*		17*		26*		36*	
1/4					9*		18*		27*		37*	
5/16							19*		28*		38*	
3/8									29*		39*	
1/2											41*	
5/8											41*	
3/4											42*	

D	NOMINAL SIZE		#10 (.190)		1/4		5/16		3/8		1/2	
B	THREADS PER INCH		32NF-3A		28UNF-3A		24UNF-3A		24UNF-3A		20UNF-3A	
J	WIDTH OF SOCKET ACROSS FLAT	MAX.	.0947		.127		.1582		.1895		.2520	
		MIN.	.0937		.125		.1562		.1875		.2500	
C	DIAMETER OF POINT	MAX.	.102		.132		.172		.212		.291	
		MIN.	.088		.118		.156		.194		.270	
T	KEY ENGAGEMENT	MIN.	.075		.100		.125		.150		.200	
L	LENGTH	TOLERANCE	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN	DASH No.	FIIN
3/16		± 1-1/2% OF L WITH A MINIMUM OF ± .010 IN.	48									
1/4			49		60		73*					
5/16			50*		61*		74		86*		98*	
3/8			51*		62		75		87		99*	
1/2			52*		63*		76*		88		100	
5/8			53*		64*		77*		89*		101*	
3/4			54*		65*		78*		90*		102*	
7/8					66*		79*		91*		103*	
1					67*		80*		92*		104*	

* INDICATES MANUFACTURER'S NON-STOCK PRODUCTION ITEMS.

KEY ENGAGEMENT: THESE DIMENSIONS APPLY TO SCREWS ONE DIAMETER IN LENGTH OR LONGER. FOR SCREWS SHORTER THAN ONE DIAMETER IN LENGTH, SOCKET TO BE AS DEEP AS PRACTICABLE.

MATERIAL: STEEL, HIGH-GRADE ALLOY, IN ACCORDANCE WITH PROCUREMENT SPECIFICATION.

HARDNESS: SETSCREWS SHALL BE HARDENED BY HEAT TREATMENT TO A HARDNESS OF ROCKWELL C45 TO 53.

PROTECTIVE COATING: PHOSPHATE COATED, IN ACCORDANCE WITH MIL-C-16232, TYPE II.

THREADS: THE THREADS SHALL BE IN ACCORDANCE WITH SCREW-THREAD STANDARDS FOR FEDERAL SERVICES HANDBOOK H-28.

DIMENSIONS: ALL DIMENSIONS ARE IN INCHES UNLESS OTHERWISE SPECIFIED.

PART NUMBER: THE MS PART NUMBER CONSISTS OF THE MS NUMBER, PLUS THE DASH NUMBER. EXAMPLE: MS51028-1.

NOTES: (1) REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.
(2) THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.

THIS DOCUMENT HAS BEEN PROMULGATED BY THE DEPARTMENT OF DEFENSE AS THE MILITARY STANDARD TO LIMIT THE SELECTION OF THE ITEM, PRODUCT, OR DESIGN COVERED HEREIN IN ENGINEERING, DESIGN, AND PROCUREMENT. THIS STANDARD SHALL BECOME EFFECTIVE NOT LATER THAN 90 DAYS AFTER THE LATEST DATE OF APPROVAL SHOWN.

REVISED
APPROVED 7 MAY 1958

CUSTODIANS	OTHER INT.
A - ORD	A-CESIGT
N - SHIPS	N-AMCOR
AF - USAF	AF-

PROCUREMENT SPECIFICATION
FF-S-103

MILITARY STANDARD

SETSCREW, HEXAGON SOCKET, FLAT POINT, ALLOY STEEL, PHOSPHATE COATED, NF-3A AND UNF-3A

SUPERSEDES:

MS51028

SHEET 1 OF 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T17:55Z
Check the source to verify that this is the current version before use.

DD FORM 1 OCT 52 672-1

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.